

Phase 1 — Understand the Problem & Define Success Banking AI Support & Advisory Agent

This phase focuses on problem framing, business understanding, user workflow analysis, success definition, and risk identification for the Banking AI Support & Advisory Agent. The purpose of this phase is to establish a strong foundation before implementing the AI system.

1. Primary User Persona

Primary User: Retail Banking Customer

The AI agent is designed primarily for customers who require informational banking assistance. These users interact with banking systems through mobile apps, websites, or customer support portals and expect quick, accurate, and safe responses.

User Characteristics: Needs fast responses for common banking queries May not understand banking terminology Requires reliable information Expects conversational interaction Needs safe handling of sensitive financial topics

2. Daily Workflow Supported

The Banking AI Agent supports the following customer workflow: User opens banking support chat. User asks a banking-related informational query. Agent identifies the intent of the request. Agent retrieves relevant banking policy information. Agent generates a safe and accurate response. If the issue is risky or ambiguous, the agent escalates the case. The interaction is logged safely without storing sensitive customer data.

3. Exact Problem Statement

Bank customers frequently contact support teams for repetitive informational questions related to loans, credit cards, EMI calculations, eligibility criteria, and policy clarification. Human support teams face scalability challenges, inconsistent response quality, and long response times.

The objective of the Banking AI Support & Advisory Agent is to: Provide instant banking support for informational queries Reduce workload on human support teams Improve response consistency and availability Operate safely under banking compliance requirements Escalate risky or unsupported situations appropriately

4. Inputs, Outputs, Constraints & Assumptions

Inputs: User banking questions Bank policy documents Conversation history User feedback **Outputs:** Banking support responses Refusal messages for unsafe requests Escalation recommendations Tool-generated calculations (e.g., EMI) **Constraints:** Cannot perform money transfers or transactions Cannot provide legal or investment advice Must not hallucinate customer information Must avoid storing PII in logs Must escalate high-risk cases **Assumptions:** Users have internet access Policy documents are updated regularly Human support escalation team is available LLM APIs are accessible during runtime

5. Example User Questions

What documents are required for a home loan? Which credit card offers travel rewards? What is the EMI for a ₹5 lakh loan? My account was charged twice. What should I do? Can you transfer ₹20,000 to another account?

6. Success Criteria

The success of the Banking AI Agent will be evaluated using the following measurable criteria:

Metric	Target
Response Accuracy	>85%
Hallucination Rate	<5%
Unsafe Request Refusal	100%
Tool Selection Accuracy	>90%
Retrieval Relevance	>80%
Average Response Latency	<5 seconds

7. Known Failure Cases & Edge Scenarios

The following risks and edge cases were identified during problem framing. These scenarios help guide future safety and engineering decisions.

Failure Case	Potential Risk	Mitigation Strategy
Hallucinated customer balance	Incorrect financial information	Refuse unsupported customer-specific queries
Money transfer request	Unauthorized transactional action	Explicit refusal guardrail
Fraud complaint	Unsafe automated handling	Human escalation workflow
Missing retrieval results	False or invented answers	Fallback response with uncertainty
Sensitive information in logs	Privacy violation	PII masking and safe logging
Ambiguous loan eligibility question	Incorrect guidance	Clarification request from user

8. Conclusion

Phase 1 established the business problem, identified the target users, defined success criteria, and documented key risks and safety requirements. This phase provides the foundation for building a reliable, explainable, and safe Banking AI Support & Advisory Agent in subsequent phases.